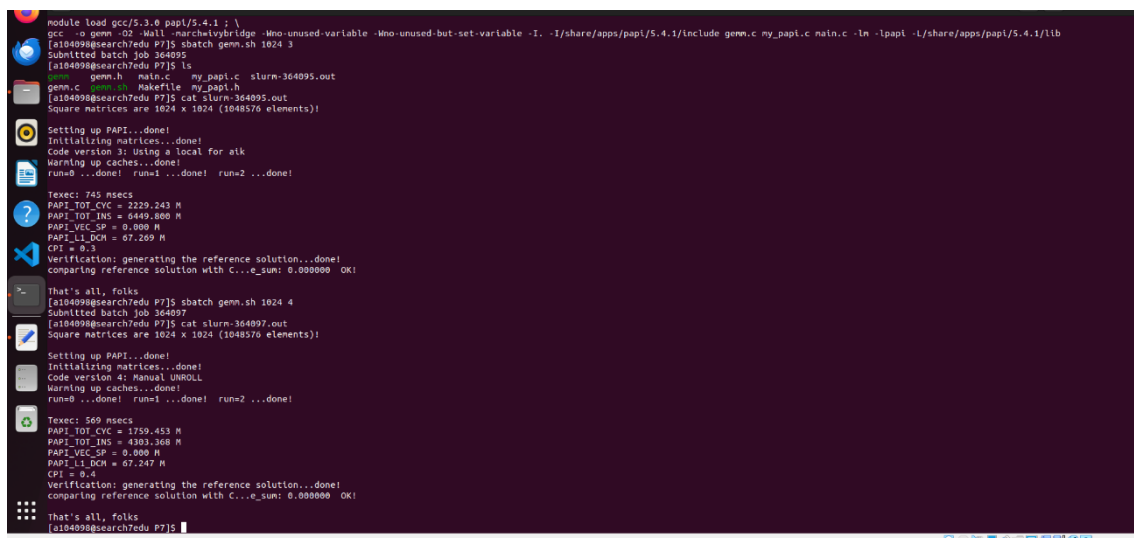


Arquitetura de Computadores

- 1- ssh a104098@s7edu.di.uminho.pt
- 2- cp /share/... exemplo: cp /share/acomp/P5-PROD.zip .
- 3- ls (opcional)
- 4- unzip ... exemplo: unzip P5-PROD.zip
- 5- ... exemplo: P5-PROD
- 6- make

outro exemplo completo (aula 14/11)

- 1- ...
- 2- nao te esqueças que é a passe que ta no email (kd3zM2qT)
- 3- cp /share/acomp/GEMM-P07.zip .
- 4- unzip GEMM-P07.zip
- 5- ls (para ver melhor)
- 6- cd P7
- 7- make
- 8- sbatch gemm.sh 1024 3 (fazer batch que provavelmente ta no enunciado)
- 9- cat slurm-364095.out
- 10- vim gemm.c (programar em vim, que é tipo vscode)
- 11- se quiser escrever/alterar o programa clico em "i"
- 12- se quiser para de escrever carrego no ESC
- 13- :wq (salvar e sair) (não esquecer que tens de ter o "ESC" pressionado)



```
module load gcc/5.3.0 papl/5.4.1 : \
gcc -o gemm -O2 -Wall -marchivybridge -Wno-unused-variable -Wno-unused-but-set-variable -I. -I/share/apps/papl/5.4.1/include gemm.c my_papl.c main.c -ln -lpapl -L/share/apps/papl/5.4.1/lib
[a104098@search7edu P7]$ sbatch gemm.sh 1024 3
Submitted batch job 364095
[a104098@search7edu P7]$ ls
gemm.h main.c my_papl.c slurm-364095.out
gemm.c Makefile my_papl.h
[a104098@search7edu P7]$ cat slurm-364095.out
Square matrices are 1024 x 1024 (1048576 elements)!

Setting up PAPI...done!
Initializing matrices...done!
Code version 3: Using a local for atk
Warning up caches...done!
run=0 ...done! run=1 ...done! run=2 ...done!

Texec: 745 msec
PAPI_TOT_CYC = 2229.243 M
PAPI_TOT_INS = 8449.800 M
PAPI_VEC_SP = 0.000 M
PAPI_VEC_DP = 0.000 M
PAPI_L1_DCH = 67.269 M
CPI = 4.3
Verification: generating the reference solution...done!
comparing reference solution with C...e_sum: 0.000000 OK!

That's all, folks
[a104098@search7edu P7]$ sbatch gemm.sh 1024 4
Submitted batch job 364097
[a104098@search7edu P7]$ cat slurm-364097.out
Square matrices are 1024 x 1024 (1048576 elements)!

Setting up PAPI...done!
Initializing matrices...done!
Code version 4: Manual Unroll
Warning up caches...done!
run=0 ...done! run=1 ...done! run=2 ...done!

Texec: 569 msec
PAPI_TOT_CYC = 1759.463 M
PAPI_TOT_INS = 4383.368 M
PAPI_VEC_SP = 0.000 M
PAPI_VEC_DP = 0.000 M
PAPI_L1_DCH = 67.247 M
CPI = 4.4
Verification: generating the reference solution...done!
comparing reference solution with C...e_sum: 0.000000 OK!

That's all, folks
[a104098@search7edu P7]$
```